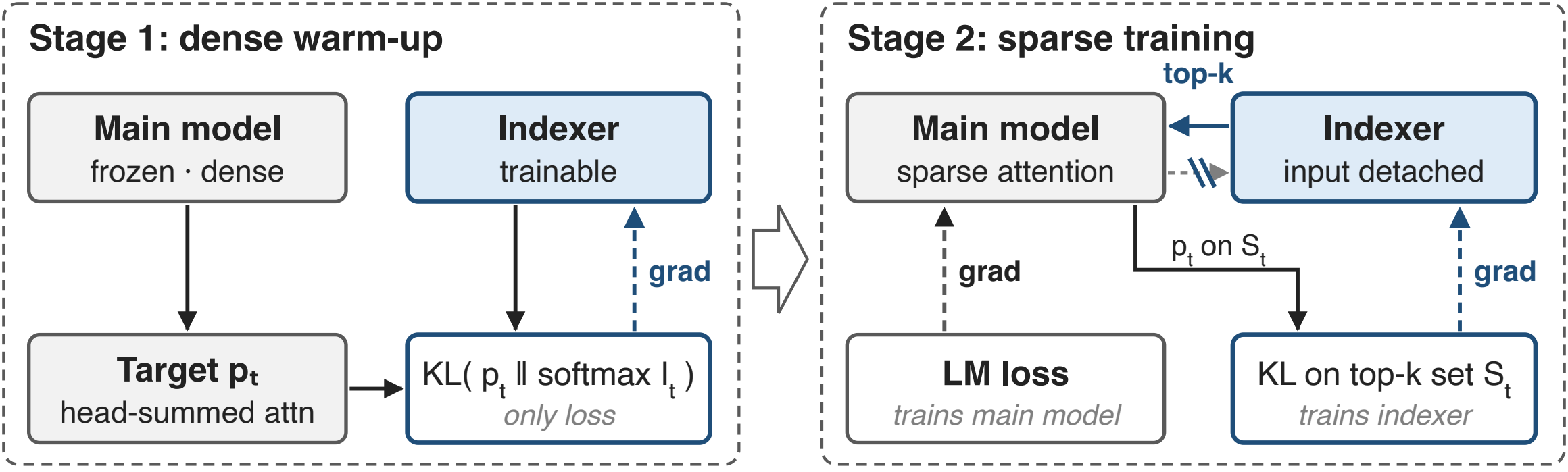

Stage 1: dense warm-up
Main model
frozen · dense
Indexer
trainable
grad
Target p_t
head-summed attn
KL(p_t ‖ softmax I_t)
only loss
Stage 2: sparse training
top-k
Main model
sparse attention
Indexer
input detached
grad
p_t on S_t
grad
LM loss
trains main model
KL on top-k set S_t
trains indexer